Prompt-Guided Cross-Network Distillation for Open-Set Cross-Domain Few-Shot Hyperspectral Image Classification

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***Abstract*—Compared with cross-domain few-shot learning (CDFSL), open-set CDFSL (OS-CDFSL) is more challenging because it must accurately classify known classes across domains while reliably rejecting unknown samples. Under this setting, domain shift causes inconsistent spectral distributions for samples from the same class and causes unknown samples to intrude into the known feature space, resulting in spectral confusion and blurred decision boundaries. Existing methods still rely on feature alignment, which may incorrectly pull unknown-class samples toward known-class regions. Moreover, threshold-based rejection strategies are sensitive to dataset characteristics and domain shifts, limiting their generalization. To address these limitations, this article proposes a prompt-guided cross-network prototype distillation (PG-CPD) method for OS-CDFSL. First, a prompt-guided foundation adaptation (PGFA) module is developed based on parameter-efficient fine-tuning (PEFT) to reduce cross-domain spectral discrepancies. Masked image modeling (MIM) optimizes learnable spatial–spectral prompts to adapt the frozen backbone to domain-specific spectral variations without using class labels. Second, a cross-network prototype distillation (CNPD) module is introduced to transfer stable semantic structures while avoiding boundary contamination from direct feature alignment. In particular, student features are evaluated with teacher semantic prototypes for cross-network logit distillation, while dynamic graph-based nonknown anchors serve as negative prototypes to form rejection regions. Finally, a threshold-free inference strategy reformulates open-set rejection as a similarity competition, avoiding manually defined thresholds. Experiments on three hyperspectral image (HSI) datasets demonstrate that the proposed method achieves state-of-the-art performance. The code is available at https://github.com/ZhengSR-031/PG-CPD**

***Index Terms*—Cross-domain few-shot learning (CDFSL), cross-network distillation, open-set hyperspectral image classification (HSIC), prompt-guided adaptation, prototype learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture ground reflectance across numerous contiguous narrow spectral bands, enabling each pixel to encode highly discriminative spatial–spectral information [1], [2]. This rich representation provides essential support for applications such as material identification [3], mineral exploration [4], environmental monitoring [5], and land-cover classification [6], [7].

Recently, deep learning models, including CNNs [8], transformers [9], and mamba-based architectures [10], have achieved remarkable success in HSI classification (HSIC) owing to their powerful feature representation capabilities. However, these models usually require large-scale annotated datasets [11] while acquiring dense pixel-level annotations in real-world scenarios is extremely labor-intensive and often impractical [12], [13]. To address this limitation, few-shot learning (FSL) has emerged as a promising paradigm for achieving reliable classification with only a limited number of labeled samples [14].

Although FSL-based methods perform well under the assumption that training and testing data follow the same distribution [15], real-world Earth observation scenarios often violate this condition [16]. HSIs acquired at different times, by different sensors, or from different geographic regions may exhibit significant variations in illumination conditions, atmospheric effects, and imaging viewpoints [17]. These domain shifts can cause dramatic degradation in model performance [18]. To address this issue, cross-domain FSL (CDFSL) attempts to transfer knowledge from a label-rich source domain to a target domain where only a few labeled samples are available [19]. For example, Shi et al. [20] enhanced cross-domain discrimination through multilevel contrastive learning,

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and Ding et al. [21] modeled global–local relations using graph attention.

However, most existing CDFSL methods adopt a closed-set episodic assumption, under which each target query sample is assumed to belong to one of the classes represented in the support set [22]. In practice, target scenes may contain categories absent from the support set, causing closed-set models to misclassify them as known classes [23]. Open-set recognition addresses this limitation, but extending it to CDFSL requires joint domain adaptation, known-class recognition, and unknown rejection under limited supervision.

Although recent methods have attempted to address open-set recognition under cross-domain few-shot settings [24], they still face several critical challenges. First, due to differences in sensors, acquisition conditions, and geographic environments, domain shifts lead to inconsistent spectral distributions of the same classes across domains, making cross-domain adaptation difficult. Second, unknown samples in the target domain may exhibit high spectral similarity with known categories but lack semantic supervision. Direct feature alignment can therefore incorrectly pull unknown samples toward known-class regions, leading to decision boundary contamination and misclassification. Third, existing rejection strategies often rely on manually selected thresholds, whose optimal values vary across datasets and domain-shift conditions, limiting their robustness and generalization. Therefore, developing a threshold-free framework that can mitigate domain discrepancies, preserve known-class discrimination, and reliably reject unknown samples under severe domain shifts and limited target annotations remains a key challenge for open-set CDFSL (OS-CDFSL).

To address these challenges, this article proposes an OS-CDFSL framework based on prompt-guided cross-network prototype distillation (PG-CPD) for HSIC. First, the prompt-guided foundation adaptation (PGFA) based on parameter-efficient fine-tuning (PEFT) is designed to reduce spectral discrepancies across domains. A spatial–spectral collaborative prompt generator produces task-specific prompt vectors to drive a masked image modeling (MIM) task on source and target data. This strategy enables the frozen foundation backbone to adapt to underlying spectral variations across domains through learnable prompts without relying on labels. Second, we introduce a cross-network prototype distillation (CNPD) to transfer stable semantic structures while avoiding the interference caused by direct feature alignment. In particular, instead of relying on low-level distance metrics or adversarial alignment with source-domain features, target features extracted by the student are fed into the teacher semantic prototype classifier to compute cross-network logits. Meanwhile, a dynamic negative prototype generator is constructed on the student side to generate nonknown negative prototypes separated from known-class positive prototypes. These prototypes do not model unknown-class semantics; instead, they form nonknown rejection spaces outside known-class regions. Finally, we design a threshold-free inference strategy that reformulates open-set recognition as a similarity competition between known-class positive prototypes and nonknown negative anchors. This strategy avoids empirical threshold tuning and improves robustness for open-set HSIC.

The main contributions of this article are summarized as follows.

1) We propose a prompt-guided foundation adaptation module for cross-domain hyperspectral scenarios. Different from conventional prompt tuning, PGFA introduces spatial–spectral collaborative prompts with masked reconstruction to capture domain-specific spectral variations without target labels.
2) We introduce a cross-network prototype distillation mechanism that avoids unreliable feature alignment in open-set scenarios. CNPD transfers teacher-guided semantic knowledge through prototype anchors and employs negative prototype generation to enhance known–unknown separation.
3) We develop a threshold-free prototype competition strategy that replaces manually selected rejection thresholds with similarity competition between known-class positive prototypes and nonknown negative anchors, improving robustness under varying domain shifts.

The remainder of this article is organized as follows. Section II reviews related work on Open-Set CDFSL for HSIC, knowledge distillation (KD), and PEFT. Section III introduces the proposed method. Section IV presents the experiments and analysis. Finally, conclusions are drawn in Section V.

II. Related Work

A. OS-CDFSL for HSIC

OS-CDFSL for HSIC aims to achieve known-class recognition and unknown-class rejection under domain shifts with limited target annotations. It is more challenging than conventional open-set HSIC due to the joint presence of domain discrepancies, few-shot supervision, and unseen categories.

Existing open-set HSIC methods mainly focus on unknown detection under single-domain settings. Prototype-based methods construct discriminative feature spaces and detect unknown samples through prototype relationships. Xie et al. [25] introduced a contrastive clustering strategy to build expandable prototypes, while Ru et al. [26] proposed progressive alignment for label-space variation. Chen et al. [27] further incorporated frequency-domain information into spectral–spatial prototypes to improve class separability.

Another category adopts generative or reconstruction mechanisms for unknown modeling. Shermin et al. [28] proposed a weighted transfer module for knowledge transfer, and Pal et al. [29] used a generative adversarial network to enhance decision boundaries. Ma et al. [30] and Yue et al. [31] explored reconstruction-based unknown detection using feature discrepancy and latent reconstruction, respectively. Liu et al. [17] further introduced extreme value theory into a multitask open-world framework for unknown recognition.

However, these methods generally assume sufficient labeled samples and consistent data distributions, limiting their applicability to realistic scenarios. Recently, HyperTaFOR [24] extended open-set recognition to CDFSL by jointly considering domain adaptation and unknown rejection. Nevertheless, existing OS-CDFSL methods still suffer from spectral discrepancies and unknown interference, while manually

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designed rejection criteria limit their generalization under domain shifts.

B. Knowledge Distillation

KD transfers knowledge from a teacher model to a student network by aligning logits or intermediate representations [32]. Benefiting from the stable supervision provided by the teacher, KD can alleviate overfitting [33] and improve efficiency and generalization in few-shot HSIC [34]. However, conventional KD struggles in open-set cross-domain scenarios [35], [36]. Because the pretrained teacher lacks target-domain adaptation, direct feature-level distillation may introduce negative transfer by pulling unknown samples toward known-class regions, while distance- or threshold-based filtering remains unreliable for open-set separation under few-shot conditions.

Inspired by semantic representation distillation [37], we introduce cross-network semantic distillation, which transfers teacher-guided class-level semantic responses rather than directly aligning feature embeddings. By constructing semantic prototypes as class anchors, the proposed strategy avoids pulling unknown samples toward known-class regions and preserves decision boundaries under open-set cross-domain few-shot scenarios.

C. Parameter-Efficient Fine-Tuning

Driven by the strong transferability of massive, large-scale pretrained foundation models across diverse downstream tasks, several remote sensing counterparts have recently emerged based on Vision transformer architectures and masked autoencoding strategies. For instance, SatMAE [38], SpectralEarth [39], and SpectralGPT [40] adopt MIM to pretrain models on large-scale multispectral imagery. More recently, HyperFree [41], built upon the HyperSigma architecture [42], enables flexible adaptation to arbitrary spectral channel numbers and diverse downstream tasks. These advances significantly enhance the generalization capability of hyperspectral models and provide a solid foundation for cross-domain learning.

Despite their strong transferability, directly fine-tuning large-scale pretrained models with limited labels often suffers from overfitting, high computational costs, and catastrophic forgetting. To address these issues, prompt engineering (PE) and PEFT have emerged as effective alternatives [43]. Instead of updating all parameters, PE adapts pretrained models through lightweight task-specific prompts while preserving the general knowledge in the backbone. In computer vision, SAM [44] further validates the potential of prompt-based adaptation for pretrained models.

Building on this paradigm, prompt-based adaptation provides an effective solution for hyperspectral foundation models under limited annotations. Instead of updating the entire backbone, lightweight prompts can adapt frozen models such as HyperFree to domain-specific spectral characteristics. When combined with masked reconstruction on unlabeled data, prompt-based strategies can capture spectral structures while preserving pretrained knowledge and alleviating overfitting under data scarcity.

III. Methods

The overall framework of the proposed PG-CPD is illustrated in Fig. 1. First, the PGFA module mitigates cross-domain distribution discrepancies by optimizing learnable prompts through prompt-driven masked reconstruction while keeping the foundation backbone frozen. Second, the CNPD module enhances discriminative representation by transferring semantic knowledge from the teacher network to the student network while constructing positive and negative prototypes to separate known-class regions from nonknown rejection regions. Finally, a threshold-free inference strategy reformulates open-set recognition as a prototype similarity competition, eliminating manually designed rejection thresholds.

A. Data Preparation

In the data preparation stage, HSIs from the source and target domains are denoted as $X_d \in \mathbb{R}^{H_d \times W_d \times B_d}$, where $d \in \{s, t\}$ denotes the source or target domain, and H_d, W_d, and B_d represent the spatial height, width, and spectral bands, respectively. Instead of densely extracting overlapping patches, we adopt a center-pixel-based sampling strategy. For each selected pixel, a local spatial–spectral cube $x_d \in \mathbb{R}^{P \times P \times B_d}$ is extracted, where P denotes the patch size. The extracted patches are generated only from the training split, while validation and test pixels are excluded to avoid spatial leakage.

To ensure consistent spectral dimensionality across domains, the first 100 valid spectral bands are retained for all datasets, resulting in $B = 100$. Each spatial–spectral patch is then projected through a linear embedding layer into a unified token representation $E \in \mathbb{R}^{N \times D}$, where $N = P^2$ and D denotes the embedding dimension. Learnable positional embeddings are further incorporated to preserve spatial structural information.

During training, samples from domain $d \in \{s, t\}$ are organized into a C-way K-shot episodic learning paradigm. The support set is defined as $\mathcal{S}_d = \{(x_i, y_i)\}_{i=1}^{C \times K}$, containing K labeled samples for each class, while the query set is defined as $\mathcal{Q}_d = \{(x_j, y_j)\}_{j=1}^{C \times Q}$, containing Q query samples per class. Query samples are randomly selected from the training split and remain disjoint from the test samples. Unknown-class samples are excluded during training and are used only for open-set evaluation.

B. Prompt-Guided Foundation Model Adaptation

Directly fine-tuning large-scale foundation models with extremely limited support samples often leads to severe overfitting and representation collapse. To achieve parameter-efficient adaptation while preserving the pretrained generalized knowledge, we propose a PGFA module. This module implicitly aligns the underlying spectral distributions of both source and target domains through a joint MIM objective driven by dynamically generated spatial–spectral collaborative prompts, where only self-supervised spectral reconstruction signals are used without category labels.

Given an extracted hyperspectral patch, the embedded token sequence is denoted as $E \in \mathbb{R}^{N \times D}$. A random masking operator

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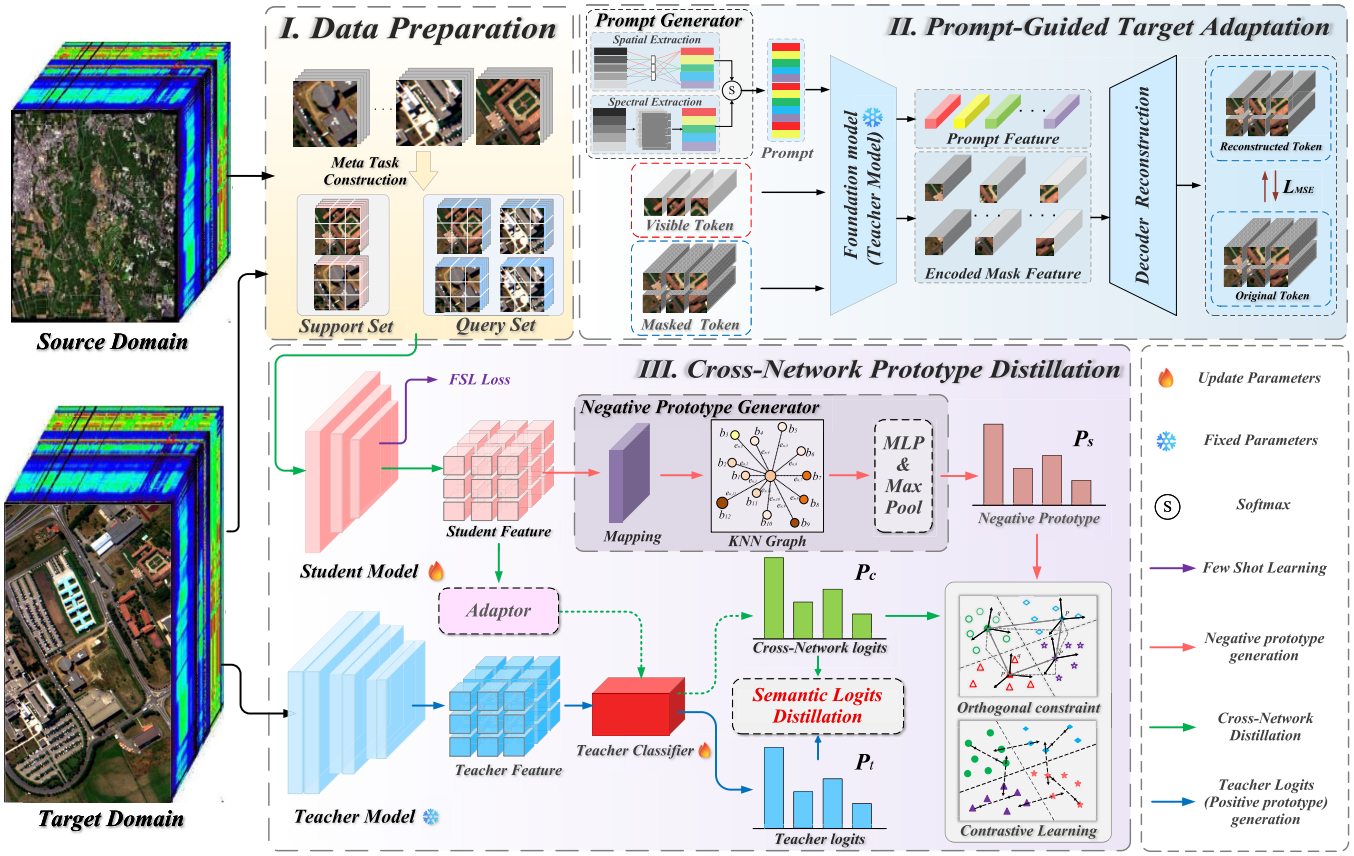


Fig. 1. Flowchart of the proposed PG-CPD method. First, source and target-domain HSI data are organized into support and query sets to construct CDFSL tasks. Then, PGFA performs prompt-guided MIM on unlabeled source and target data to alleviate cross-domain shifts. Finally, CNPD transfers teacher-guided semantic structures and generates known-class positive prototypes and nonknown negative anchors as negative prototypes, enabling threshold-free open-set classification through prototype similarity competition. (a) Data preparation. (b) Prompt-guided target adaptation. (c) Cross-network prototype distillation.

$\mathcal{M}(\cdot)$ is applied to E with a masking ratio γ, which divides the token sequence into visible tokens $E_v \in \mathbb{R}^{N_v \times D}$ and masked tokens $E_m \in \mathbb{R}^{N_m \times D}$. In particular, $E_v, E_m = \mathcal{M}(E, \gamma)$, where $N_v = \lfloor (1-\gamma)N \rfloor$ and $N_m = N - N_v$. Learnable positional embeddings are further incorporated into the token sequence to preserve the spatial structural information of hyperspectral patches.

1) Spatial–Spectral Collaborative Prompt Generation: Directly projecting flattened patches into prompt tokens entangles spatial topology with continuous spectral signatures, thereby weakening the physical interpretability of hyperspectral representations. To address this issue, we design a dual-branch prompt generator that explicitly decouples and jointly models spatial and spectral cues across domains.

Given visible tokens E_v and learnable queries $Q \in \mathbb{R}^{L \times D}$, the generator extracts shared representations via cross-attention and generates spectral- and spatial-aware prompts through two lightweight branches to obtain spectral- and spatial-aware candidate prompts

$$P_{\text{spec}} = \mathcal{F}_{\text{spec}}(Q, E_v), \quad P_{\text{spa}} = \mathcal{F}_{\text{spa}}(Q, E_v). \tag{1}$$

To adaptively balance spatial and spectral information across cross-domain instances, a softmax-based gating mechanism is introduced to dynamically fuse the two branch outputs into the final prompt sequence P

$$P = \alpha \odot P_{\text{spec}} + \beta \odot P_{\text{spa}} \tag{2}$$

where $[\alpha, \beta] = \text{Softmax}(W_g[P_{\text{spec}} \parallel P_{\text{spa}}])$, W_g is a learnable gating matrix, $\odot$ denotes elementwise multiplication, and $[\alpha, \beta] \in \mathbb{R}^{L \times 2}$ are the normalized fusion weights satisfying $\alpha + \beta = 1$. This dynamic softmax-normalized fusion enables the frozen backbone to selectively emphasize the most discriminative spatial–spectral characteristics of each input sample while preserving its pretrained generalized knowledge.

2) Parameter-Efficient Fine-Tuning: In this work, we adopt the pretrained hyperspectral foundation model HyperFree [41] as the backbone. To achieve PEFT under limited samples, the generated prompt vectors P are prepended to the visible tokens, forming the input sequence H_0

$$H_0 = [P \parallel E_v] \in \mathbb{R}^{(L+N_v) \times D} \tag{3}$$

where $[\cdot \parallel \cdot]$ denotes concatenation along the sequence dimension, L denotes the prompt length, N_v is the number of visible tokens, and D is the embedding dimension. This prompt-injected sequence is subsequently fed into the strictly frozen HyperFree backbone.

During forward propagation, the prompt tokens interact with visible HSI tokens through the frozen transformer blocks.

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After the final layer, the prompt tokens are removed, and the remaining visible representations are denoted as H_{out}^{v}.

To facilitate label-free spectral adaptation, we further introduce a lightweight decoder D_ϕ, which is optimized using only self-supervised reconstruction signals without category labels. A set of learnable mask tokens $M \in \mathbb{R}^{N_m \times D}$ is introduced to replace the masked regions. These mask tokens are concatenated with the encoded visible representation H_{out}^{v} and restored to their 2-D spatial topology to preserve structural consistency.

The decoder D_ϕ then operates on the reconstructed full token sequence to predict the normalized spectral signatures of the masked patches. The entire PGFA module is optimized by minimizing the mean squared error reconstruction loss $\mathcal{L}_{\text{mse}}$ computed only over the masked regions

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N_m} \sum_{i \in \mathcal{M}} \left\| x_i - \hat{x}_i \right\|_2^2 \tag{4}$$

where $\hat{x} = D_\phi(\text{Restore}(H_{out}^{v}, M))$, x_i and $\hat{x}_i$ denote the original and reconstructed spectral values of the ith masked patch for domain d, respectively, and N_m represents the number of masked tokens. The operator Restore$(\cdot)$ reassembles visible and mask tokens into their original layout before decoding. During backpropagation, only the prompt generator and decoder are updated. No class-level supervision is involved during this adaptation stage. After PGFA optimization, the decoder is discarded.

C. Cross-Network Prototype Distillation

Conventional feature alignment may cause unknown samples to be misaligned with known classes. To address this issue, we propose a CNPD module. In particular, CNPD exploits few-shot support information to construct semantic prototypes, transfers stable knowledge through cross-network distillation, and introduces positive and negative prototypes with contrastive learning to separate known-class regions from nonknown rejection regions.

1) Cross-Domain FSL: In open-set cross-domain few-shot scenarios, the teacher and student play different roles; therefore, we adopt an asymmetric teacher–student architecture. The teacher network F_t combines the adapted prompt generator with the frozen HyperFree backbone to provide stable semantic guidance, while the student network F_s adopts a trainable ViT backbone to learn target-oriented representations and construct prototypes through episodic learning. This design avoids overfitting caused by directly fine-tuning foundation models with scarce labels while enabling effective knowledge transfer from stable semantic priors to task-specific representations.

Within each episode, all samples are projected by the student network F_s into a shared embedding space. For class k, its prototype is defined as the mean feature of its support samples

$$c_k^d = \frac{1}{\left|\mathcal{S}_{d,k}\right|} \sum_{x_i \in \mathcal{S}_{d,k}} F_s\left(x_i\right) \tag{5}$$

where $\mathcal{S}_{d,k}$ denotes the subset of $\mathcal{S}_d$ labeled as class k.

Query samples are classified by measuring their distances to these prototypes. The probability that a query sample $x_j \in \mathcal{Q}_d$ belongs to class k is computed as

$$P\left(y = k \mid x_j\right) = \frac{\exp\left(-G_E\left(F_s\left(x_j\right), c_k^d\right)\right)}{\sum_{r=1}^{C} \exp\left(-G_E\left(F_s\left(x_j\right), c_r^d\right)\right)} \tag{6}$$

where $G_E(\cdot,\cdot)$ denotes the Euclidean distance, and y_j represents the ground-truth label of x_j. By optimizing this prototype-based classification objective, the student network learns compact and discriminative class representations around their corresponding prototypes, thereby improving cross-domain generalization.

The episodic FSL loss for a specific domain is computed by minimizing the negative log-likelihood of the ground-truth labels over its respective query set instances. Formally, for a given domain indicator $d \in \{s, t\}$, the domain-specific FSL loss function is defined as

$$\mathcal{L}_{\text{fsl}}^{d} = \frac{\sum_{(x_j, y_j) \in \mathcal{Q}_d} -\log P\left(y = y_j \mid x_j\right)}{\left|\mathcal{Q}_d\right|} \tag{7}$$

where $|\mathcal{Q}_d|$ denotes the number of query samples in domain d.

By jointly optimizing both domains, the student network learns domain-invariant spectral representations while mitigating target-domain overfitting. The cross-domain FSL loss is

$$\mathcal{L}_{\text{fsl}} = \mathcal{L}_{\text{fsl}}^{s} + \mathcal{L}_{\text{fsl}}^{t}. \tag{8}$$

2) Cross-Network Semantic Distillation: Although the FSL strategy provides basic discriminability, directly aligning teacher and student features may impose rigid feature overlap and weaken open-set separability. To address this issue, we introduce a cross-network semantic distillation mechanism that transfers semantic responses from the frozen teacher F_t to the student network. The teacher parameters are not updated using target category labels; labeled target support samples are used only to construct episodic teacher semantic prototypes for semantic distillation.

The teacher semantic prototypes are defined as $P_t = \{\mu_t^1, \ldots, \mu_t^C\} \in \mathbb{R}^{C \times D_t}$, where μ_t^k denotes the centroid of teacher-side support features for the kth known class. For a target query sample $x_j \in \mathcal{Q}_t$, the teacher feature is obtained as $f_t = F_t(x_j) \in \mathbb{R}^{D_t}$. Meanwhile, the student ViT outputs token features $F_s(x_j) \in \mathbb{R}^{N \times D_s}$, which are aggregated into $f_s \in \mathbb{R}^{D_s}$ via global average pooling. A lightweight MLP adapter $\mathcal{A}(\cdot)$ with linear projection, GELU activation, and LayerNorm then maps f_s into the teacher semantic space as $\tilde{f}_s = \mathcal{A}(f_s)$ for cross-network semantic distillation.

The teacher predictive distribution is computed by measuring the cosine similarity between the L_2-normalized teacher feature f_t and the teacher semantic prototypes P_t

$$p_t^k = \frac{\exp\left(\text{sim}\left(f_t, \mu_t^k\right)/\tau\right)}{\sum_{c=1}^{C} \exp\left(\text{sim}\left(f_t, \mu_t^c\right)/\tau\right)}. \tag{9}$$

Meanwhile, the cross-network predictive distribution is computed by evaluating the L_2-normalized adapted student

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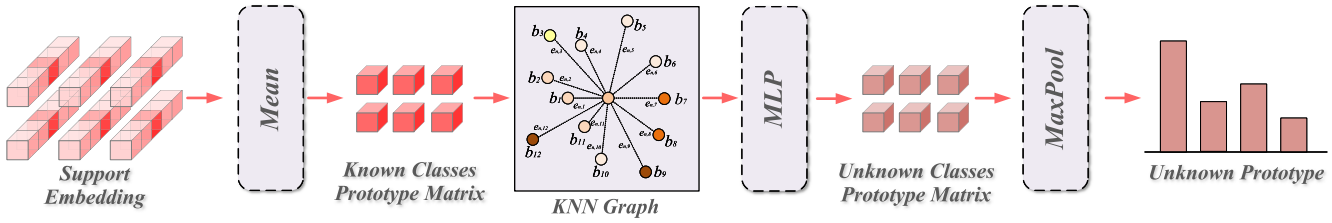


Fig. 2. Schematic of the negative prototype generation module. Learnable negative anchors retrieve their k nearest adapted support embeddings and are transformed into task-adaptive negative prototypes through residual edge aggregation.

representation $\tilde{f}_s$ with the same teacher semantic prototype set

$$p_c^k = \frac{\exp\left(\text{sim}\left(\tilde{f}_s, \mu_t^k\right)/\tau\right)}{\sum_{c=1}^{C}\exp\left(\text{sim}\left(\tilde{f}_s, \mu_t^c\right)/\tau\right)} \tag{10}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ is the temperature parameter.

To transfer the teacher-defined semantic response structure, we minimize the Kullback–Leibler divergence between the teacher predictive distribution and the cross-network predictive distribution

$$\mathcal{L}_{\text{KL}} = \frac{1}{|\mathcal{Q}_t|}\sum_{x_j\in\mathcal{Q}_t}\sum_{k=1}^{C} p_t^k \log\frac{p_t^k}{p_c^k}. \tag{11}$$

3) Negative Prototype Generation: In open-set few-shot scenarios, unknown-classes lack semantic supervision and should not be explicitly modeled by generated prototypes. Therefore, we construct nonknown negative anchors around known-class neighborhoods to represent rejection regions near decision boundaries rather than specific unknown-class semantics. In particular, a k-nearest neighbor (k-NN) graph with residual edge transformation is introduced to encode anchor-neighbor offsets and capture boundary-sensitive variations. As illustrated in Fig. 2, NPG jointly generates known-class positive prototypes and nonknown negative anchors from the support set $\mathcal{S}_t$.

The positive prototypes $P^+ = \{p_1^+, \ldots, p_C^+\}$ are computed as centroids of student support features. To construct nonknown rejection regions, we introduce U learnable negative anchors $b_n \in \mathbb{R}^D$, $(n = 1, \ldots, U)$. Given the adapted student feature embeddings of the support set, denoted as $\tilde{f}_s$, a k-NN graph is constructed based on cosine similarity between negative anchors and support features.

For each anchor b_n, its k nearest support features are aggregated via a residual edge transformation to capture boundary-aware nonknown representations

$$e_{(n,m)} = \text{MLP}\left(\left[b_n \,\|\, \left(\tilde{f}_{n,m}^s - b_n\right)\right]\right) \tag{12}$$

where $[\cdot \,\|\, \cdot]$ denotes feature concatenation. The nth unknown negative prototype is then obtained by symmetric max pooling over the neighborhood

$$p_n^- = \max_{m=1,\ldots,k} e_{(n,m)} \tag{13}$$

where the max operation is performed elementwise over the k neighboring edge features. The learnable anchor b_n is thus transformed into a task-adaptive negative prototype p_n^-. These negative prototypes do not represent unknown-class semantics but construct adaptive nonknown rejection regions based on the local geometry of known support features.

To stabilize the generated negative anchors and prevent them from becoming task-agnostic noisy vectors, we introduce a local geometry regularization term

$$\mathcal{L}_{\text{geo}} = \frac{1}{kU}\sum_{n=1}^{U}\sum_{m=1}^{k}\|\tilde{f}_{n,m}^s - p_n^-\|_2^2. \tag{14}$$

No additional gradient clipping or normalization is applied specifically to $\mathcal{L}_{\text{geo}}$, as it only preserves the local neighborhood structure of negative prototypes rather than determining their final positions. Since it does not explicitly enforce separation from known-class prototypes, especially for spectrally correlated classes, the subsequent contrastive prototype learning objective further optimizes their relative arrangement to prevent prototype collapse.

4) Contrastive Prototype Learning: To address this issue, we introduce a contrastive prototype learning strategy to separate known-class regions from nonknown rejection regions. Student-side support centroids serve as known-class positive prototypes P^+, while the generated nonknown anchors form negative prototypes P^-. The augmented prototype space is then defined as $P_{\text{aug}} = P^+ \cup P^-$, where P^+ contains C known-class positive prototypes and P^- contains U nonknown negative prototypes.

To prevent the generated negative prototypes from collapsing into known-class positive prototypes, we impose a normalized cosine orthogonality constraint

$$\mathcal{L}_{\text{oc}} = \frac{1}{CU}\sum_{c=1}^{C}\sum_{n=1}^{U}\left(\frac{\left(p_c^+\right)^\top p_n^-}{\|p_c^+\|_2\|p_n^-\|_2}\right)^2. \tag{15}$$

We further introduce a contrastive objective in the prototype metric space. For a target query sample x_j with episodic label $y_j = c$, its adapted representation $\tilde{f}_{s,j} = \mathcal{A}(F_s(x_j))$ is encouraged to approach the corresponding positive prototype while being repelled from other positive prototypes and all negative prototypes. The objective follows a supervised InfoNCE formulation:

$$\mathcal{L}_{\text{con}} = -\frac{1}{N}\sum_{j=1}^{N}\log\frac{\exp\left(\text{sim}\left(\tilde{f}_{s,j}, p_{y_j}^+\right)/\tau\right)}{\sum_{p\in P_{\text{aug}}}\exp\left(\text{sim}\left(\tilde{f}_{s,j}, p\right)/\tau\right)} \tag{16}$$

where $N = |\mathcal{Q}_t|$, $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ is the temperature parameter.

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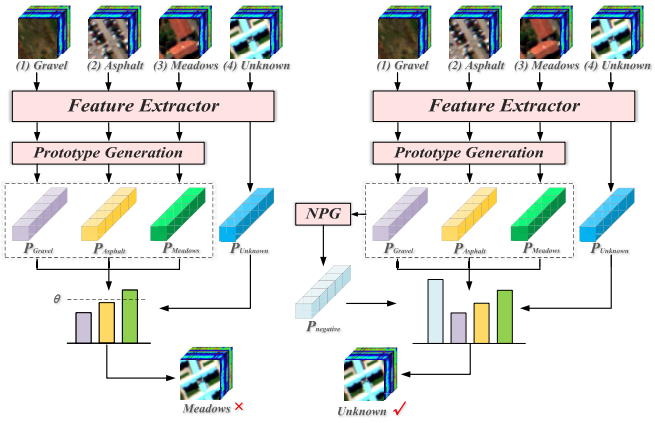


Fig. 3. Comparison between conventional threshold-based methods and the proposed method.

The loss function of contrastive prototype learning can be expressed as

$$\mathcal{L}_{\mathrm{CPL}} = \mathcal{L}_{\mathrm{oc}} + \mathcal{L}_{\mathrm{con}}. \tag{17}$$

No fixed margin or manually specified boundary interval is imposed between positive and negative prototypes; instead, their relative separation is adaptively learned through $\mathcal{L}_{\mathrm{oc}}$ and $\mathcal{L}_{\mathrm{con}}$. Accordingly, the overall CNPD loss integrates FSL, semantic distillation, negative prototype generation, and contrastive prototype learning as follows:

$$\mathcal{L}_{\mathrm{CNPD}} = \mathcal{L}_{\mathrm{fsl}} + \lambda_1 \mathcal{L}_{\mathrm{KL}} + \lambda_2 \mathcal{L}_{\mathrm{geo}} + \lambda_3 \mathcal{L}_{\mathrm{CPL}} \tag{18}$$

where λ_1, λ_2, and λ_3 are positive learnable weights initialized to 1 and jointly optimized through backpropagation to balance semantic distillation, negative prototype generation, and contrastive prototype learning.

D. Threshold-Free Open-Set Inference

As illustrated in Fig. 3, the proposed framework adopts a lightweight student-only inference paradigm. The teacher network and reconstruction decoder are discarded after training, leaving only the student network, adapter, negative prototype generator, and support-derived prototype classifier for efficient open-set prediction.

For an unseen target sample x_t, the student backbone F_s together with the adapter $\mathcal{A}$ extracts the adapted representation $\tilde{f}_t = \mathcal{A}(F_s(x_t))$. The feature is then evaluated in the augmented prototype space P_{aug} to obtain the joint prediction distribution

$$p_{\mathrm{joint}}^{(k)} = \frac{\exp\left(\mathrm{sim}\left(\tilde{f}_t, p_{\mathrm{aug}}^{(k)}\right)/\tau\right)}{\sum_{c=1}^{C+U} \exp\left(\mathrm{sim}\left(\tilde{f}_t, p_{\mathrm{aug}}^{(c)}\right)/\tau\right)}. \tag{19}$$

The final prediction is determined by selecting the maximum probability over the augmented label space

$$\hat{y} = \arg \max_{j \in \{1,\ldots,C+U\}} p_{\mathrm{joint}}^{(j)}. \tag{20}$$

If $\hat{y} \le C$, the sample is classified as the $\hat{y}$th known category; otherwise it is assigned to the unknown class. This formulation converts open-set recognition into a standard $(C + U)$-way classification problem, avoiding manually tuned sample-level rejection thresholds.

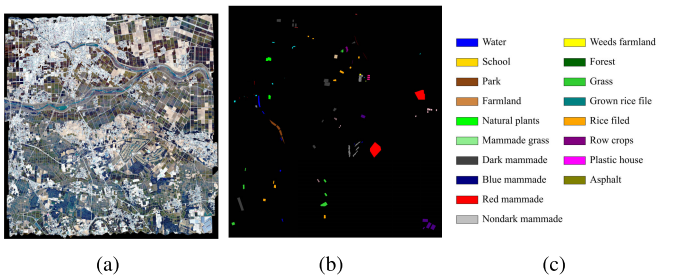


Fig. 4. Visualization of the source dataset. (a) Pseudocolor composite images. (b) Ground truth maps. (c) Corresponding color labels.

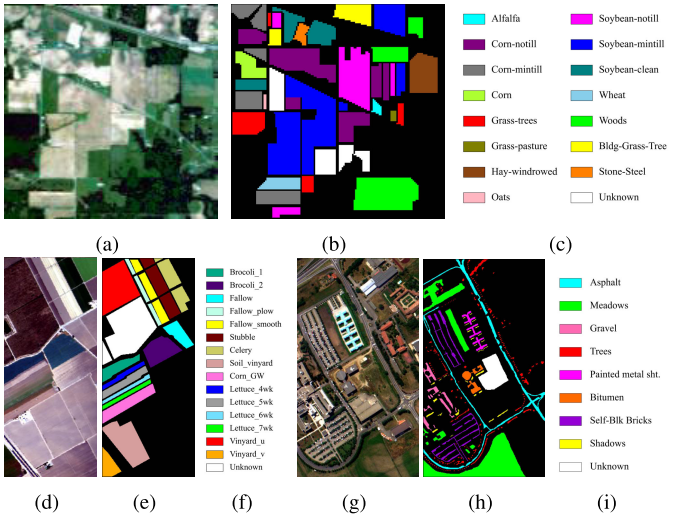


Fig. 5. Visualization of the three target datasets. (a), (d), and (g) Pseudocolor composite images. (b), (e), and (h) Ground truth maps. (c), (f), and (i) Corresponding color labels.

IV. Experiments

A. Datasets

Figs. 4 and 5 show pseudocolor images, ground-truth maps, and corresponding color labels. The land-cover classes and the number of samples for each class are presented in Table I.

1) Source-Domain Dataset: The Chikusei dataset was acquired by the Headwall Hyperspec-VNIR-CIRIS sensor over a complex agricultural–urban landscape in Ibaraki, Japan. The image covers a spatial extent of 2517 × 2335 pixels with a spatial resolution of 2.5 m. Spectrally, it contains 128 contiguous bands ranging from 363 to 1018 nm. The dataset is annotated with 19 land-cover categories.

2) Target-Domain Datasets: The Indian Pines (IPs) dataset was collected by the AVIRIS sensor over an agricultural and forested region in Indiana. It contains 145 × 145 pixels with a spatial resolution of 20 m. After removing 24 bands affected by water absorption, 200 spectral bands are retained. The scene includes 16 classes, representing different growth stages and canopy conditions of corn and soybean crops.

The Salinas (SAs) dataset was captured by the AVIRIS sensor over an agricultural area in California, USA. The image size is 512 × 217 pixels and originally contains 224 spectral bands spanning the visible and near-infrared ranges. It includes 16 agricultural classes, such as different growth stages of romaine lettuce, vineyards, and fallow soils.

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TABLE I
LAND-COVER CLASSES AND NUMBERS OF SAMPLES IN THE SOURCE AND TARGET DATASETS

NO.	Chikusei		Indian Pines (IP)		Salinas (SA)		Pavia University (PU)	
	Class Name	Total	Class Name	Total	Class Name	Total	Class Name	Total
1	Water	2845	Alfalfa	46	Brocoli_1	2009	Asphalt	6631
2	Bare soil (school)	2859	Corn-notill	1428	Brocoli_2	3726	Meadows	18649
3	Bare soil (park)	286	Corn-mintill	830	Fallow	1976	Gravel	2099
4	Bare soil (farmland)	48525	Corn	237	Fallow_plow	1394	Trees	3064
5	Natural plants	4297	Grass-trees	730	Fallow_smooth	2678	Painted metal sht.	1345
6	Weeds in farmland	1108	Grass-pasture-mowed	28	Stubble	3959	Bitumen	1330
7	Forest	20516	Hay-windrowed	478	Celery	3579	Self-Blk Bricks	3682
8	Grass	6515	Oats	20	Soil_vinyard	6203	Shadows	947
9	Rice field (grown)	13369	Soybean-notill	972	Corn_GW	3278	**Bare_Soil(Unk.)**	**5029**
10	Rice field (first stage)	1268	Soybean-mintill	2455	Lettuce_4wk	1068		
11	Row crops	5961	Soybean-clean	593	Lettuce_5wk	1927		
12	Plastic house	2193	Wheat	205	Lettuce_6wk	916		
13	Manmade (non-dark)	1220	Woods	1265	Lettuce_7wk	1070		
14	Manmade (dark)	7664	Bldg-Grass-Tree	386	Vinyard_u	7268		
15	Manmade (blue)	431	Stone-Steel	93	Vinyard_v	1807		
16	Manmade (red)	222	**Grass-pasture (Unk.)**	**483**	**Grapes_untr. (Unk.)**	**11271**		
17	Manmade grass	1040						
18	Asphalt	801						
19	Paved ground	145						
	Total	**121265**	**Total**	**10249**	**Total**	**54129**	**Total**	**42776**

The Pavia University (PU) dataset was acquired by the ROSIS-03 sensor over an urban campus in Italy. The scene has spatial dimensions of 610×340 pixels. After removing noisy bands affected by water vapor absorption, 103 spectral bands remain. The ground truth consists of nine urban categories, including asphalt, meadows, gravel, and bare soil.

B. Experimental Details

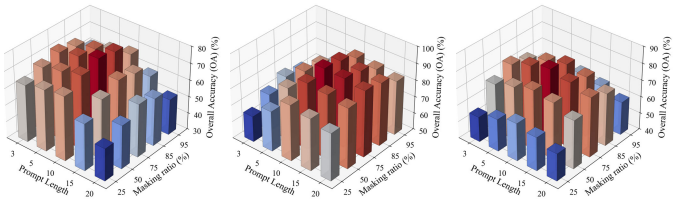
All experiments are conducted on an Intel Core Ultra7 CPU, 64 GB RAM, and an NVIDIA GeForce RTX 4090 GPU using Python 3.11 and PyTorch 2.3.1. We evaluate the performance using OA, AA, Kappa, and open-set metrics, including known accuracy, unknown recall, and H-score. The H-score is defined as

$$H = \frac{2A_{\mathrm{k}}R_{\mathrm{unk}}}{A_{\mathrm{k}} + R_{\mathrm{unk}}} \tag{21}$$

where A_{k} and R_{unk} denote the known accuracy and unknown recall, respectively. All results are averaged over 10 independent runs. All HSI samples are represented as 9×9 spatial–spectral blocks. To ensure spectral dimensionality unification across domains, the spectral channels are uniformly fixed to $B = 100$ by retaining the first 100 valid spectral bands.

During feature adaptation, HyperFree-B is adopted as the foundation backbone and kept frozen, with only prompt-related parameters and the reconstruction module optimized. PGFA uses only unlabeled samples from the source and target training splits. Target test samples and all unknown-class samples are excluded from both PGFA and CNPD training. The model is trained for 5000 episodes with an initial learning rate of 0.001 and a weight decay of 1×10^{-4}.

The CDFSL task follows a standard C-way K-shot episodic setting. In the target domain, five labeled samples per known class form the available labeled pool. For each episode, $K = 1$, with one sample randomly selected from these five labeled samples as support and 19 query samples generated through label-preserving augmentation. Thus, the 5-shot setting refers to the number of available labeled samples per class rather than the episodic support size. During evaluation, the same labeled pool is used to construct known-class support sets, while unknown-class samples are introduced only for open-set testing.



(a) (b) (c)

Fig. 6. Hyperparameter analysis of the masking ratio and prompt length on three datasets. (a) IP dataset. (b) SA dataset. (c) PU dataset.

C. Parameter Analysis

For the prompt-guided foundation adaptation stage, we investigate the effects of prompt length and random masking ratio, as shown in Fig. 6. The prompt length controls the adaptation capacity of the frozen backbone, while the masking ratio determines the reconstruction difficulty. The results show that a prompt length of ten and a masking ratio of 75% achieve the best performance.

Then, we analyze the temperature coefficient τ in the distillation, as shown in Fig. 7. A smaller τ produces sharper teacher–student response distributions, while a larger τ generates smoother semantic distributions. The optimal τ varies across datasets, where IP and SA achieve the best performance at $\tau = 0.2$, while PU obtains the best result at $\tau = 0.1$.

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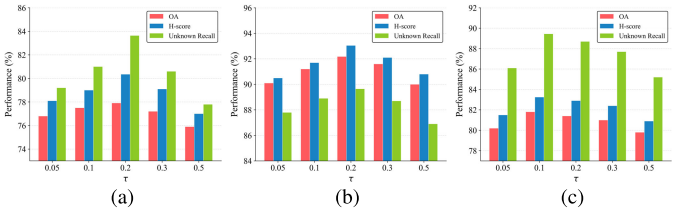


Fig. 7. Sensitivity analysis of the temperature coefficient τ on three datasets. (a) IP dataset. (b) SA dataset. (c) PU dataset.

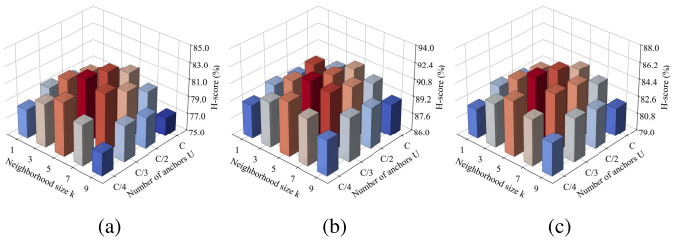


Fig. 8. Hyperparameter analysis of the number of neighbors k and the number of negative prototypes U on three datasets. (a) IP dataset. (b) SA dataset. (c) PU dataset.

Furthermore, we analyze the effects of the neighborhood size k and the number of negative prototypes U, as shown in Fig. 8. The parameter k controls the spatial–spectral neighborhood scale used for negative prototype construction. A small k captures fine-grained local variations but may be sensitive to few-shot noise, whereas a large k provides broader coverage but may weaken boundary specificity. The parameter U controls the density and coverage of the constructed rejection regions. Similarly, too few negative prototypes lead to insufficient rejection-space coverage, while too many introduce redundant constraints. The best performance is achieved at $k = 5$ and $U = C/3$, providing a favorable balance between known-class recognition and unknown-class rejection.

D. Comparative Experiments

1) Comparison With State-of-the-Art Methods: To evaluate the effectiveness of PG-CPD, we compare it with representative closed-set HSIC, CDFSL, open-set HSIC, and open-set CDFSL methods. In our OS-CDFSL setting, the Chikusei dataset is utilized as the source domain, while IP, SA, and PU are adopted as target domains. For closed-set HSIC methods such as SSRN [45] and CDFSL methods including Gia-CFSL [46] and DAPKD-CFSL [34], we adapt them to the open-set setting using validation-based rejection thresholds. In particular, samples whose maximum prediction confidence is lower than the validation-selected threshold are assigned to the unknown class. We also compare with dedicated open-set HSIC methods, including the semantic-reconstruction-based HyperCASR [47] and the prototype-based FrFSSPN [27], as well as the recent task-adaptive open-set CDFSL method HyperTaFOR [24]. All methods are evaluated under the same known/unknown splits.

For each target-domain task, five labeled samples per known class are provided as the annotation budget, while the remaining target samples are used for testing. Unknown-class samples are only involved in the final open-set evaluation. Each method is independently run ten times under identical settings, and the average results are reported. The Unknown row denotes unknown recall, and H-score is reported to evaluate the balance between known-class recognition and unknown-class rejection. The best performance for each metric is highlighted in bold.

Table II presents the classification performance of different methods on the IP dataset. The proposed PG-CPD improves OA, AA, and Kappa by 2.19%, 3.43%, and 8.53% over the second-best results, respectively, and achieves the highest H-score of 82.51%. DAPKD-CFSL obtains competitive known-class recognition but suffers from poor unknown rejection with only 46.16% unknown recall, which indicates strong closed-set overfitting. Although other open-set methods improve unknown rejection, they often degrade known-class discrimination. On the contrary, PG-CPD achieves an unknown recall of 83.64% while maintaining strong known-class recognition, demonstrating a better balance between known and unknown categories.

The results on the SA dataset are reported in Table III. Since SA contains more land-cover categories and complex spectral variations, it poses greater challenges for both known-class discrimination and unknown-sample rejection. PG-CPD achieves the best OA, AA, and Kappa values of 92.18%, 91.11%, and 90.76%, respectively. It also obtains the highest H-score of 91.81% and an unknown recall of 92.53%, outperforming HyperTaFOR by 4.83% in unknown rejection. These results demonstrate the effectiveness of positive–negative prototype competition in improving known–unknown separation under complex spectral variations.

Table IV reports the results on the PU dataset. PG-CPD achieves the highest OA, AA, and Kappa values of 81.81%, 84.52%, and 80.74%, respectively, together with the best H-score of 85.94% and unknown recall of 87.41%. These consistent improvements over DAPKD-CFSL and HyperTaFOR demonstrate that PG-CPD achieves a better balance between known-class recognition and unknown-sample rejection in open-set cross-domain few-shot HSIC.

For intuitive visual comparison, Figs. 9–11 illustrate the classification maps generated by different methods on the three datasets, corresponding to the quantitative results reported in Tables II–IV. The ground-truth maps are also provided for reference. As shown in Fig. 9, on the IP dataset, the classification map produced by PG-CPD exhibits the highest spatial consistency with the ground truth and accurately identifies the unknown regions. On the SA dataset (see Fig. 10), a small number of unknown samples (white) are occasionally misclassified as the red-labeled class, which is mainly attributed to the high spectral similarity between unknown and known categories. A similar phenomenon can also be observed on the PU dataset (see Fig. 11), where some unknown pixels (white) are mistakenly assigned to the green-labeled class due to the spectral resemblance between unknown and known classes. Despite these spectral ambiguities, PG-CPD still produces the most visually consistent classification results, demonstrating its strong robustness in distinguishing known classes while effectively rejecting unknown samples.

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE IP DATASET

Class	SSRN [46]	Gia-CFSL [47]	DAPKD-CFSL [35]	HyperCASR [48]	FrFSSPN [27]	HyperTaFOR [24]	Ours
1	41.81±4.21	**98.81±3.52**	94.59±2.14	94.57±4.13	94.59±3.62	93.41±2.95	96.85±1.85
2	66.38±5.11	45.01±6.24	67.64±5.47	46.07±4.83	46.78±5.15	51.16±5.25	**70.86±4.38**
3	21.79±7.23	44.73±8.15	47.82±6.25	45.69±7.14	46.18±6.82	**52.48±4.27**	50.27±5.93
4	27.30±6.15	**79.55±5.42**	77.21±4.35	74.22±6.28	74.70±5.84	75.43±4.16	78.22±3.82
5	82.45±3.45	75.65±4.53	81.41±3.86	75.29±5.17	75.96±4.92	79.00±4.29	**83.15±3.15**
6	88.12±4.18	**89.48±5.27**	84.96±4.15	82.40±6.34	82.82±5.46	83.32±4.12	87.24±3.56
7	36.33±8.52	98.39±2.18	97.60±1.53	98.99±3.85	**99.40±3.26**	97.42±0.94	97.50±2.47
8	87.41±3.68	**99.85±1.13**	98.46±2.05	85.36±4.58	85.79±4.14	92.29±3.52	96.19±2.83
9	62.11±5.24	98.12±1.26	99.59±0.03	98.04±2.87	**99.50±1.15**	98.87±0.82	98.80±1.18
10	59.64±4.85	52.02±5.14	66.47±4.37	68.95±6.16	70.37±5.82	**72.90±4.56**	65.87±3.64
11	71.96±5.52	64.28±6.37	71.34±5.19	62.34±7.25	63.89±6.54	67.04±5.28	**73.44±4.87**
12	51.20±7.15	51.04±8.23	51.18±6.46	44.90±9.17	45.96±8.54	47.77±7.58	**53.61±5.26**
13	87.49±3.82	98.35±1.56	99.15±0.84	96.46±3.28	98.14±2.85	**99.59±1.24**	97.81±1.16
14	**97.95±2.14**	81.91±4.57	95.39±3.83	77.92±5.48	78.32±4.95	83.04±4.24	97.56±3.17
15	38.01±8.25	73.01±5.46	70.94±4.65	67.05±6.53	67.20±5.86	68.63±4.84	**73.82±3.57**
Unknown	37.01±8.15	48.69±6.52	46.16±5.83	72.15±4.26	77.90±3.93	82.46±2.56	**83.64±1.85**
OA	61.75±4.24	67.28±3.74	75.45±3.16	71.80±3.54	73.15±3.25	75.72±2.86	**77.91±1.50**
AA	59.81±6.87	74.93±2.69	78.12±2.45	74.40±3.14	75.50±2.82	77.80±2.57	**81.55±1.85**
Kappa	56.97±7.63	58.89±3.46	64.34±3.18	60.10±3.52	61.80±3.27	65.40±2.84	**73.93±1.67**
H-score	46.16±6.26	59.56±4.05	58.61±3.43	73.33±3.85	76.58±4.32	79.90±3.32	**82.51±2.74**

TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE SA DATASET

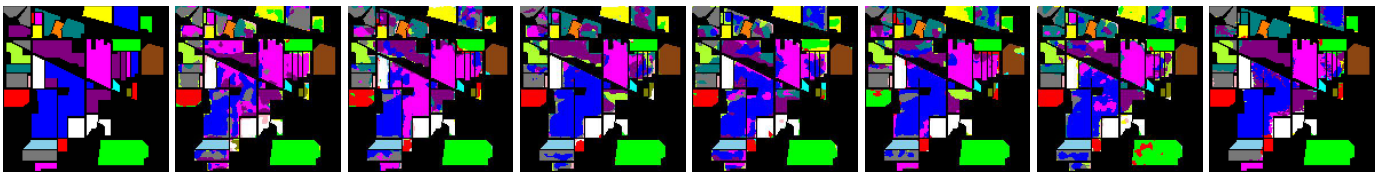
Class	SSRN [46]	Gia-CFSL [47]	DAPKD-CFSL [35]	HyperCASR [48]	FrFSSPN [27]	HyperTaFOR [24]	Ours
1	80.11±5.70	85.34±4.81	89.21±2.14	88.06±4.35	88.73±3.40	92.81±2.14	**94.50±1.85**
2	83.39±4.66	85.71±6.90	93.26±3.30	85.32±5.16	86.41±4.23	91.93±3.96	**94.55±2.91**
3	80.50±5.12	87.67±4.24	90.65±1.66	90.04±3.98	90.71±3.19	**94.67±2.42**	93.71±1.36
4	82.78±3.28	88.20±2.22	86.03±1.96	91.71±2.78	92.12±2.21	**98.28±1.13**	94.01±1.36
5	77.99±4.72	88.77±1.65	88.33±0.66	89.13±1.92	89.67±1.80	**92.20±1.91**	92.11±1.72
6	81.11±4.67	89.35±4.63	96.72±2.97	88.36±4.29	89.42±3.58	95.21±2.22	**98.67±1.46**
7	72.16±5.48	83.46±3.61	84.90±4.04	73.34±5.24	74.83±4.70	68.58±5.86	**89.63±2.75**
8	62.19±6.77	73.96±7.50	**93.25±3.57**	70.12±6.86	82.57±6.01	83.74±5.44	84.67±4.51
9	66.28±7.02	72.35±6.48	84.94±4.49	72.77±5.47	73.98±4.96	84.21±3.36	**89.80±2.85**
10	63.74±6.38	72.73±5.96	81.52±5.80	76.23±6.13	77.86±5.98	**85.91±4.99**	85.43±2.21
11	69.54±8.69	77.15±5.35	83.84±4.43	80.01±5.03	80.69±4.75	**93.72±2.61**	87.97±3.48
12	56.26±5.07	62.67±6.25	83.37±5.32	67.14±6.47	68.98±5.54	**86.33±4.92**	80.67±3.81
13	77.76±5.20	85.73±3.72	85.97±1.83	89.53±3.47	90.96±2.36	**95.11±1.65**	93.81±0.94
14	96.95±4.21	97.43±1.54	98.24±0.20	98.72±1.91	95.11±1.20	95.21±0.80	**98.89±0.40**
15	79.15±2.50	83.77±0.82	82.69±0.10	84.94±1.55	86.13±1.17	**88.48±0.51**	86.81±0.20
Unknown	59.58±7.11	60.23±6.20	59.32±3.17	72.98±5.20	78.23±4.80	87.70±3.50	**92.53±2.15**
OA	73.54±3.44	78.45±2.45	84.15±2.11	81.50±2.49	84.10±2.21	89.87±1.84	**92.18±0.95**
AA	74.12±2.81	80.20±2.39	86.39±1.92	82.40±2.22	84.15±2.05	90.56±1.52	**91.11±0.88**
Kappa	69.85±3.63	74.60±2.57	81.20±2.19	78.30±2.60	80.10±2.40	88.40±1.93	**90.76±0.92**
H-score	66.06±5.75	68.80±4.59	70.34±3.20	77.40±3.29	81.08±3.05	89.11±2.37	**91.81±1.89**

TABLE IV
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE PU DATASET

Class	SSRN [46]	Gia-CFSL [47]	DAPKD-CFSL [35]	HyperCASR [48]	FrFSSPN [27]	HyperTaFOR [24]	Ours
1	68.23±5.12	83.93±4.15	**90.12±3.14**	82.16±4.13	83.11±3.67	87.53±2.95	89.58±1.85
2	70.04±4.82	61.61±6.25	**76.47±5.41**	70.26±4.83	68.29±5.15	72.88±4.28	75.65±3.36
3	75.39±6.27	91.64±5.14	**94.94±2.19**	85.64±6.17	87.84±5.83	92.68±4.95	93.08±1.45
4	80.16±4.15	87.67±3.45	88.79±4.37	81.40±5.26	84.02±4.89	86.94±3.12	**89.15±2.84**
5	85.55±3.46	96.26±2.58	**98.65±1.15**	88.96±4.18	91.47±3.91	96.12±2.23	96.79±0.95
6	70.65±4.19	85.37±5.27	86.70±4.15	78.86±5.32	81.08±4.46	84.93±4.11	**88.40±3.54**
7	55.56±7.56	66.71±6.14	72.24±5.51	62.61±7.87	64.45±6.28	68.96±5.49	**75.21±4.93**
8	42.48±8.25	55.84±7.69	59.66±6.13	49.02±8.55	51.04±7.17	55.90±6.52	**65.41±5.86**
Unknown	36.06±7.15	44.18±6.57	47.93±5.82	65.30±5.24	71.81±4.92	80.35±3.58	**87.41±1.85**
OA	65.40±4.24	68.50±3.74	77.30±3.16	72.80±3.55	73.50±3.22	78.40±2.87	**81.81±1.13**
AA	64.90±5.87	74.80±2.69	79.50±2.42	73.80±3.17	75.90±2.83	80.70±2.56	**84.52±1.45**
Kappa	61.20±6.63	66.40±3.46	70.80±3.14	68.50±3.52	70.10±3.28	75.20±2.81	**80.74±1.25**
H-score	46.34±6.83	55.55±5.94	59.86±4.79	69.27±4.84	73.80±4.12	80.52±3.68	**85.94±2.17**

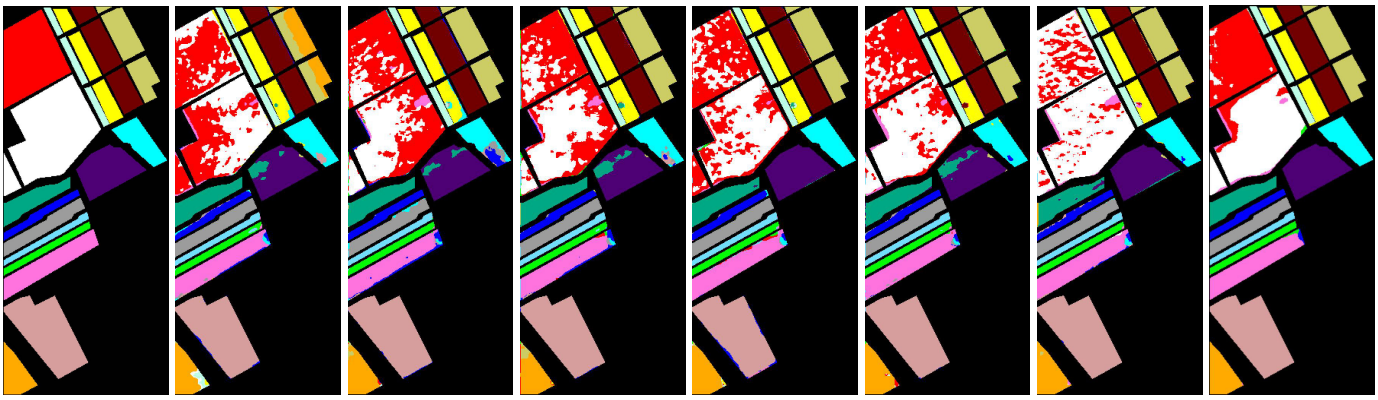
2) Computational Complexity: To further evaluate computational complexity, Table V reports the training and testing time, FLOPs, params, and peak GPU memory usage. FLOPs and GPU memory are reported in the format of

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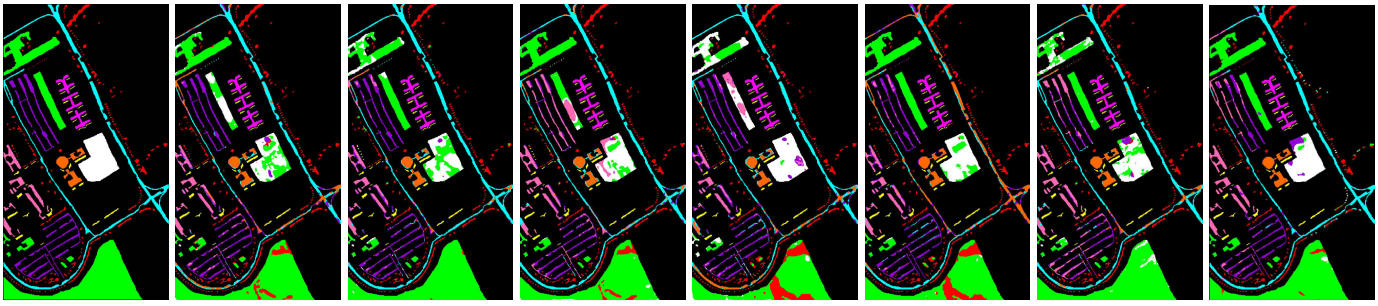
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 9. Classification maps for the IP dataset. (a) Ground-truth. (b) SSRN. (c) Gia-CFSL. (d) DAPKD-CFSL. (e) HyperCASR. (f) FrFSSPN. (g) HyperTaFOR. (h) Ours.



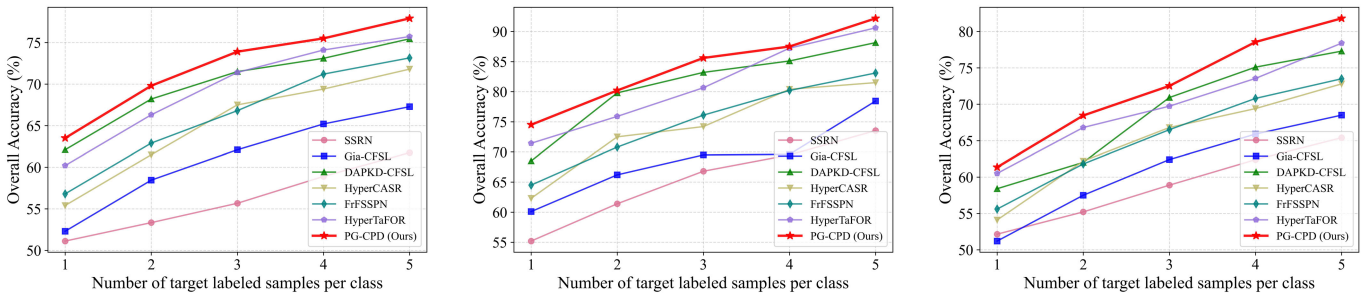
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 10. Classification maps for the SA dataset. (a) Ground truth. (b) SSRN. (c) Gia-CFSL. (d) DAPKD-CFSL. (e) HyperCASR. (f) FrFSSPN. (g) HyperTaFOR. (h) Ours.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 11. Classification maps for the PU dataset. (a) Ground truth. (b) SSRN. (c) Gia-CFSL. (d) DAPKD-CFSL. (e) HyperCASR. (f) FrFSSPN. (g) HyperTaFOR. (h) Ours.



(a) (b) (c)

Fig. 12. Effect of the number of labeled training samples per class on the overall accuracy in different datasets: (a) IP, (b) SA, and (c) PU.

training/inference, whereas params are reported as trainable/deployed params. For PG-CPD, the training cost includes the PGFA preadaptation and CNPD stages. Separately, the HyperFree backbone contains approximately 93 M parameters. Of these, 92.24 M parameters are frozen and therefore excluded from the trainable parameter count. The inference

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TABLE V
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS

Dataset	Metrics	SSRN [46]	Gia-CFSL [47]	DAPKD-CFSL [35]	HyperCASR [48]	FrFSSPN [27]	HyperTaFOR [24]	Ours
IP	Training Time (s)	825.40	2605.00	2360.10	2150.30	2245.80	2850.40	3150.80
	Testing Time (s)	3.50	9.90	8.20	6.30	6.80	10.50	9.40
	FLOPs (G)	1.84/1.84	2.60/2.60	3.61/2.32	2.15/2.15	2.25/2.25	2.46/2.46	4.75/1.49
	Params (M)	1.25/1.25	4.85/4.85	10.24/4.24	3.95/3.95	4.13/4.13	4.56/4.56	6.42/2.85
	GPU Memory (GB)	5.16/4.82	5.45/5.15	9.84/4.56	10.45/9.45	11.58/10.98	10.92/10.02	10.04/4.84
SA	Training Time (s)	1245.50	3605.00	3008.00	2950.50	3123.60	3857.20	3965.40
	Testing Time (s)	5.20	13.00	8.60	9.50	12.80	14.50	12.10
	FLOPs (G)	1.84/1.84	2.60/2.60	3.61/2.32	2.15/2.15	2.25/2.25	2.46/2.46	4.75/1.49
	Params (M)	1.25/1.25	4.85/4.85	10.24/4.24	3.95/3.95	4.13/4.13	4.56/4.56	6.42/2.85
	GPU Memory (GB)	5.28/5.14	6.83/6.34	10.84/4.95	11.45/6.45	12.08/8.58	11.53/10.77	11.24/5.35
PU	Training Time (s)	750.20	3034.20	2178.50	1950.80	2050.40	2245.50	2650.80
	Testing Time (s)	3.20	5.90	5.80	5.50	6.10	6.20	5.60
	FLOPs (G)	1.84/1.84	2.60/2.60	3.61/2.32	2.15/2.15	2.25/2.25	2.46/2.46	4.75/1.49
	Params (M)	1.25/1.25	4.85/4.85	10.24/4.24	3.95/3.95	4.13/4.13	4.56/4.56	6.42/2.85
	GPU Memory (GB)	4.85/4.45	5.18/5.02	9.88/4.68	10.08/9.48	11.61/10.61	10.92/9.79	9.98/4.68

TABLE VI
FRAMEWORK-LEVEL ABLATION STUDY ON THREE DATASETS

Metric	PGFA	CNPD	IP	SA	PU
Unknown	×	×	35.24±3.15	38.52±3.40	32.35±4.10
	✓	×	50.35±3.02	55.60±3.15	46.45±3.80
	×	✓	75.28±2.16	82.36±2.42	76.85±2.38
	✓	✓	**83.64±1.85**	**92.53±2.15**	**87.41±1.85**
Known	×	×	74.51±2.31	78.36±2.10	70.40±2.45
	✓	×	75.31±2.10	86.24±1.90	80.65±2.15
	×	✓	74.15±1.48	85.69±1.18	78.67±1.28
	✓	✓	**81.41±2.38**	**91.10±1.79**	**84.52±2.08**
H-score	×	×	47.85±2.45	51.65±2.70	44.33±2.80
	✓	×	60.35±2.60	67.61±2.20	58.95±2.60
	×	✓	74.71±1.90	83.99±1.50	77.75±1.70
	✓	✓	**82.51±2.74**	**91.81±1.89**	**85.94±2.17**

TABLE VII
ABLATION STUDY OF PGFA ON THREE DATASETS

Metric	Method	IP	SA	PU
Unknown	HyperFree	77.35±2.86	86.42±3.12	81.25±2.74
	+MIM	79.12±2.45	88.35±2.85	83.18±2.52
	+MIM+VPT	81.24±2.15	90.35±2.28	84.82±2.05
	+MIM+Adapter	81.56±2.05	89.84±2.42	85.14±1.98
	+MIM+SSPG (Ours)	**83.64±1.85**	**92.53±2.15**	**87.41±1.85**
Known	HyperFree	78.50±2.25	88.48±1.65	80.49±1.84
	+MIM	79.20±1.92	89.19±1.42	81.50±1.55
	+MIM+VPT	80.50±1.65	90.09±1.25	83.20±1.38
	+MIM+Adapter	80.19±1.58	90.50±1.18	83.79±1.32
	+MIM+SSPG (Ours)	**81.41±2.38**	**91.10±1.79**	**84.52±2.08**
H-score	HyperFree	77.92±3.20	87.44±3.42	80.87±3.86
	+MIM	79.16±2.85	88.77±2.85	82.33±3.12
	+MIM+VPT	80.87±2.45	90.22±2.34	84.00±2.62
	+MIM+Adapter	80.87±2.38	90.17±2.15	84.46±2.35
	+MIM+SSPG (Ours)	**82.51±2.74**	**91.81±1.89**	**85.94±2.17**

complexity considers the deployed student network, adapter, NPG, and prototype-based inference operations. The slightly increased testing time is mainly caused by prototype construction and negative prototype generation, which enhance known–unknown separation. Although PG-CPD introduces additional training overhead, it achieves lower inference FLOPs and competitive parameter and GPU memory costs compared with most baselines.

3) Different Numbers of Training Samples: To further evaluate PG-CPD under different annotation budgets, we varied the number of labeled training samples in the target domain. In particular, we randomly selected from one to five annotated samples per class and repeated the experiments five times. The OA of various methods across the three datasets is illustrated in Fig. 12. As shown in Fig. 12, the OA generally improves as more labeled samples become available. PG-CPD consistently outperforms the compared methods under all annotation settings, demonstrating its robustness to limited target supervision.

E. Ablation Experiment

1) Framework-Level Ablation Experiment: To evaluate the effectiveness of each component in PG-CPD, we conduct a comprehensive ablation study under the five-sample annotation budget. The experiments include framework-level and module-level analyses to verify the contribution of each component. Table VI presents the framework-level ablation results. Compared with the baseline, PGFA improves both unknown rejection and overall classification performance by reducing cross-domain spectral discrepancy, while CNPD achieves larger gains by enhancing known–unknown discrimination through semantic transfer and prototype-based rejection modeling. Combining PGFA and CNPD achieves the best performance across all datasets, demonstrating their complementary effects.

2) Ablation Experiment for PGFA: To further analyze PGFA, Table VII illustrates the comparison of different foundation model adaptation strategies. Without adaptation, the frozen HyperFree backbone provides a strong initialization but suffers from target-domain spectral discrepancy. MIM reduces this discrepancy by learning shared spatial–spectral structures, while SSPG further improves adaptation by generating spatial–spectral collaborative prompts. Compared with conventional PEFT strategies such as VPT and adapters, SSPG

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TABLE VIII
ABLATION STUDY OF CNPD ON THREE DATASETS

Metric	Method	IP	SA	PU
Unknown	Feature KD	63.24±2.72	66.35±2.86	60.42±2.95
	CSD	64.50±2.50	72.15±2.65	62.60±2.95
	CSD+Global Neg.	75.42±2.45	78.54±2.38	77.85±2.52
	CSD+NPG w/o $\mathcal{L}_{\text{geo}}$	74.15±2.32	78.21±2.45	77.32±2.48
	CSD+NPG	77.28±2.16	85.36±2.42	80.85±2.38
	CSD+NPG+CPL	**83.64±1.85**	**92.53±2.15**	**87.41±1.85**
Known	Feature KD	70.90±1.75	84.47±1.42	80.80±1.55
	CSD	77.37±1.52	85.69±1.35	83.49±1.40
	CSD+Global Neg.	78.54±1.45	79.10±1.32	79.81±1.38
	CSD+NPG w/o $\mathcal{L}_{\text{geo}}$	80.34±1.52	87.18±1.25	78.79±1.42
	CSD+NPG	79.78±1.48	90.70±1.18	81.67±1.28
	CSD+NPG+CPL	**81.41±2.38**	**91.10±1.79**	**84.52±2.08**
H-score	Feature KD	66.85±2.40	74.32±2.25	69.14±2.45
	CSD	70.35±1.60	78.34±1.42	71.55±1.45
	CSD+Global Neg.	76.95±1.48	78.82±1.35	78.82±1.36
	CSD+NPG w/o $\mathcal{L}_{\text{geo}}$	77.12±1.55	82.45±1.42	78.05±1.40
	CSD+NPG	78.51±1.54	87.95±1.25	81.26±1.35
	CSD+NPG+CPL	**82.51±2.74**	**91.81±1.89**	**85.94±2.17**

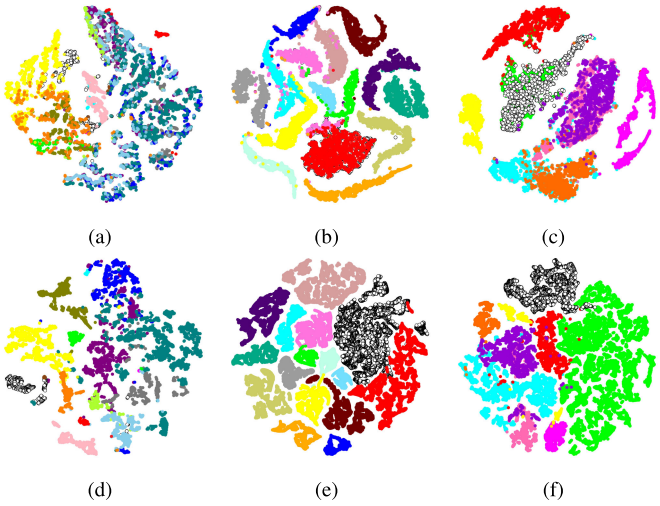


Fig. 13. t-SNE visualization of feature distributions on the IP, SA, and PU datasets. (a)–(c) Correspond to the raw spectral feature space, whereas (d)–(f) depict the feature space learned by PG-CPD. The unknown class is represented by white markers with black edges.

achieves superior performance, demonstrating its effectiveness for hyperspectral foundation model adaptation.

3) Ablation Experiment for CNPD: Table VIII evaluates different CNPD variants. Compared with feature distillation, CSD improves performance by transferring semantic responses. Compared with global negative prototypes, NPG achieves better performance by exploiting k-NN local neighborhoods to generate boundary-aware nonknown prototypes around known-class regions. Removing the geometric regularization $\mathcal{L}_{\text{geo}}$ degrades performance, indicating that it preserves local structural relationships and prevents ineffective negative prototypes. Finally, CPL further improves known–unknown separation by constraining the relative positions of positive and negative prototypes through prototype competition.

4) Feature Distribution Visualization: To intuitively assess the discriminative capability of the learned representations, Fig. 13 shows the comparison of the raw spectral features with those learned by PG-CPD. The raw features exhibit substantial overlap between known and unknown categories, whereas PG-CPD forms more compact known-class clusters and clearer known–unknown separation. This demonstrates that PG-CPD learns a more discriminative open-set feature space, thereby improving unknown-class rejection.

V. CONCLUSION

In this article, we proposed a PG-CPD for open-set cross-domain few-shot HSIC. The proposed framework addresses the key challenges caused by severe domain shift, limited target-domain annotations, and unknown-class intrusion, which jointly lead to spectral confusion and blurred decision boundaries. In particular, the PGFA module mitigates cross-domain spectral discrepancies through prompt-guided masked reconstruction while keeping the foundation backbone frozen. The CNPD module enhances known–unknown discrimination by transferring stable semantic knowledge via cross-network logit distillation and constructing nonknown negative prototypes for rejection region modeling. In addition, the threshold-free inference strategy avoids manually tuned rejection thresholds and improves decision robustness. Experimental results on three benchmark datasets demonstrate that PG-CPD consistently outperforms existing state-of-the-art methods. Since PG-CPD introduces additional training costs from PGFA and cross-network distillation, future work will focus on developing more lightweight prompt adaptation and efficient knowledge transfer strategies. We will also extend the framework to more generalized open-set HSIC scenarios.

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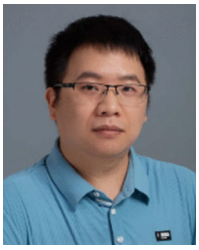
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